Planning the Next Evidence Acquisition Step: An Offline Framework for Active Scientific Discovery

Scientific Evidence Planner Project
Paper 4 — offline evidence planning for active scientific discovery

August 2026

Abstract

Most work on machine-assisted science focuses on generating scientific questions or executing experiments autonomously. We address the step in between: given an existing scientific question, a set of competing hypotheses, and the evidence currently available, *what should we investigate next to maximize expected scientific progress?* We present an offline, domain-independent planning framework that represents evidence as a typed, confidence-weighted graph; identifies hypotheses with missing, weak, or non-independent evidence; generates generic candidate evidence-acquisition actions; and ranks them by a configurable utility combining expected information gain, novelty, tractability, and cost. Beliefs over hypotheses are revised with a modular Bayesian updater, closing the plan–observe–update loop without ever executing actions. Because the planner is offline, it can be evaluated honestly against history: a benchmark truncates the evidence graph at a time cutoff, hides later evidence as ground truth, and asks whether the recommended actions would have resolved the question earlier. On a worked case study from exoplanet atmospheric characterization — a reported strongly subsolar water abundance whose premise was later refuted by three independent re-analyses — the planner's top recommendation is exactly the discriminating re-analysis that historically resolved the question (resolution efficiency 1.0; replaying the refuting evidence removes 1.06 bits of hypothesis entropy). We further ground the utility components in the historical-outcome findings of the companion predictor-discovery study (Paper 3, n=980 questions at five cutoffs): explicitly articulated competing hypotheses predict resolution; the resolution channel runs through quantified, explicitly contradictory, recurrent evidence rather than methodological doubt; and semantic novelty does not predict uptake — findings we convert into an empirically calibrated weight configuration that sharpens the planner's ranking toward the historically resolving actions. Scaling the benchmark to all 374 of Paper 3's tension questions — each evidence cluster batch-converted to a graph by transparent rules and planned at its own historical cutoff — turns the metrics into distributions and surfaces a structural result: planner utility is systematically *higher* on questions the community subsequently neglected (permutation $p \approx 0.004$), quantifying the divergence between resolution-driven and attention-driven allocation of scientific effort. The framework is released as a reference implementation with seven evaluation metrics, full configurability, and no domain assumptions in code.

1 Introduction

Scientific progress is rarely limited by the ability to ask questions; it is limited by the cost of acquiring the evidence that discriminates between competing answers. A researcher facing an open question — is a reported effect real, an artifact of the analysis, a confounder, or an instrument systematic? — must decide where to spend the next unit of effort: re-read the literature, re-run an analysis with different assumptions, compare independent datasets, run a simulation, or invest in a costly new observation. This decision is made under uncertainty, under resource constraints, and usually informally.

This paper treats that decision as a planning problem. We assume the scientific question already exists (produced by a human or by an upstream question-generation system) and ask only: *given the current evidence, which evidence-acquisition step maximizes expected scientific progress?*

This is Paper 4 of a series. Paper 1 built an evidence-graph system that generates questions from cross-paper observational tensions; Paper 2 introduced historical backtesting of question generators (freeze questions on pre-cutoff literature, observe what the community actually did afterwards); Paper 3[1] scaled that protocol into a supervised predictor-discovery study over 980 astronomy questions frozen at five cutoffs (2012–2020) on the complete arXiv astro-ph record, with strict blinded outcome labels. Those papers ask which *questions* the future rewards; this paper takes the question as given and plans the *evidence acquisition* that follows. Paper 3's outcome dataset feeds back into this work in two concrete ways developed in Section 4: its controlled effects empirically ground (and partly correct) our utility components, and its released questions and outcome labels are directly importable as benchmark units for our evaluation protocol.

Our design commitments differ from much recent work on autonomous science:

- **Offline.** The system is a planner, not an agent. It does not browse, does not call instrument APIs, and does not execute experiments. Everything works from static datasets, which makes the system auditable and makes rigorous historical evaluation possible.
- **Domain-independent.** No scientific field is hardcoded. Domain knowledge enters only through data (questions, hypotheses, evidence graphs) and configuration (action catalogs, weights).
- **Modular.** Every component — hypothesis generation, gap analysis, action generation, each utility term, belief updating, evaluation — sits behind a small interface and can be replaced independently.

Contributions. (1) A formalization of next-step evidence planning over typed, confidence-weighted evidence graphs with provenance; (2) a gap analysis that flags hypotheses with missing, sparse, weak, or non-independent evidence; (3) a configurable utility decomposition — expected information gain via preposterior analysis, provenance-based novelty, resource-aware tractability, and cost; (4) a historical-simulation benchmark with seven metrics that evaluates whether a planner would have resolved a question earlier; (5) an empirical grounding of the utility components in Paper 3's historical-outcome dataset, including an import path for its released questions and labels and an empirically calibrated weight configuration; (6) a scaled benchmark run over all 374 of Paper 3's tension questions, batch-converted to evidence graphs by auditable rules, with metric distributions and an outcome-stratified analysis showing that planner utility anti-correlates with subsequent community attention; and (7) an open reference implementation with unit tests and a worked case study whose hidden ground truth follows the documented historical outcome.

2 Problem Statement

Let q be a scientific question and $\mathcal{H} = \{h_1, \ldots, h_n\}$ a set of mutually exclusive, exhaustive competing hypotheses with probabilities $p = (p_1, \ldots, p_n)$, $\sum_i p_i = 1$. Let $G = (V, E)$ be the current *evidence*

[1] https://github.com/nonameisready/scientific-question-outcome-prediction-indicators

graph (Section 3.1). Let $\mathcal{A}$ be a set of candidate evidence-acquisition actions, each with a type, relative cost, relative time, required resources, and a target hypothesis.

The planner must return a ranking of $\mathcal{A}$ (top-K recommendations) maximizing a utility

$$U(a) \;=\; w_{\mathrm{ig}}\,\mathrm{IG}(a) \;+\; w_{\mathrm{nov}}\,\mathrm{Nov}(a) \;+\; w_{\mathrm{tr}}\,\mathrm{Tr}(a) \;-\; w_c\,C(a), \tag{1}$$

where IG is the expected reduction in hypothesis uncertainty, Nov measures independence from existing evidence, Tr measures feasibility, C is cost, and the weights w are configurable. After an action is (externally) executed and its outcome observed, hypothesis probabilities are revised and the cycle repeats. The framework plans and updates; it never executes.

3 Framework

3.1 Evidence Graphs and Provenance

Evidence is a directed multigraph with five node types — *paper*, *observation*, *dataset*, *claim*, *hypothesis* — and four edge types — *supports*, *contradicts*, *derived_from*, *observes* — each edge carrying a confidence in $[0, 1]$. Claims point at the hypotheses they support or contradict; derived nodes point at their origins (claim → paper → dataset); observations point at what they observe. Nodes optionally carry timestamps, which enables historical simulation (Section 5).

Two queries are load-bearing. First, evidence(h) returns the supports/contradicts edges incident to a hypothesis. Second, roots(v) follows *derived_from* edges transitively to the root sources of a node. Two claims that descend from the same dataset are *not* independent evidence, no matter how many papers separate them; this notion of provenance drives both gap analysis and novelty.

3.2 Hypothesis Generation

Hypotheses are structured objects (identifier, description, prior, supporting and contradicting evidence references). The reference implementation provides two interchangeable generators behind one interface: a *static* generator that loads curated hypothesis sets keyed by question (the normal path for imported datasets), and a *template* generator producing four generic methodological alternatives applicable to any empirical claim: the effect is real; it is an artifact of the analysis method; it is confounded by an unmodeled factor; it is an instrument or measurement systematic. Priors are normalized, defaulting to uniform.

3.3 Evidence Gap Analysis

For each hypothesis the analyzer emits the first applicable gap:

1. **No evidence.** No edge touches the hypothesis (gap confidence 1.0).
2. **Sparse evidence.** Fewer than $k_{\min}$ evidence edges; confidence $0.5 + 0.5\,(1 - \mathrm{count}/k_{\min})$.
3. **Weak support.** Summed support confidence below a threshold $s_{\min}$; confidence $0.4 + 0.4\,\delta$ where δ is the relative deficit.
4. **No independent corroboration.** All evidence descends from a single provenance root (confidence 0.78): formally, $\left|\bigcup_{v\in\mathrm{sources}(h)} \mathrm{roots}(v)\right| \le 1$.

The output is a gap report: hypothesis, missing-evidence description, confidence, and a human-readable reason, sorted most-confident first. All thresholds are configurable.

3.4 Candidate Action Generation

Actions are instantiated from a generic, YAML-configurable catalog of seven types: read paper, search archive, re-run analysis, collect observation, run simulation, compare datasets, search review. Each template carries a relative cost and time in $[0, 1]$ and required resource tags. For each gap, the generator instantiates the subset of types that plausibly closes that kind of gap — independence gaps receive independent-source actions (collect observation, compare datasets, search archive); weak-support gaps receive re-analysis actions; evidence-free hypotheses receive broad search actions — and records the gap's reason and confidence in the action's metadata. Nothing in the catalog or the mapping references any scientific domain.

3.5 Utility Estimation

Each term of Eq. (1) is an independent estimator behind a common interface, injected into the combiner; any term can be replaced without touching the others.

Expected information gain (preposterior analysis). An action a targeting hypothesis h_t is modeled as producing a binary outcome $o \in \{\text{favors } h_t,\ \text{disfavors } h_t\}$ with a likelihood ratio r_a determined by the action type's *discriminative power* (a configurable table; an independent observation separates hypotheses more sharply than reading a paper):

$$P(o^+ \mid h_t) = \frac{r_a}{r_a + 1}, \qquad P(o^+ \mid h_j) = \frac{1}{r_a + 1} \quad (j \neq t). \tag{2}$$

The estimate is the expected entropy reduction

$$\mathrm{IG}(a) \;=\; H(p) \;-\; \mathbb{E}_o\big[H\big(p \mid o\big)\big] \;\geq\; 0, \tag{3}$$

with H the Shannon entropy in bits and the posterior computed by Bayes' rule for each outcome. This is exact for the two-outcome model and degrades gracefully: actions with likelihood ratio 1 yield zero gain, as do actions with no target among the live hypotheses.

Novelty (provenance independence). Each action type has a base source novelty $\nu_a \in [0, 1]$ (new observation $= 1.0$; re-reading literature ≈ 0.3). The base is boosted by the provenance concentration of the target's existing evidence:

$$\mathrm{Nov}(a) \;=\; \min\Big(\nu_a\big(1 + \tfrac{1}{\max(|R_t|, 1)}\big),\; 1\Big), \tag{4}$$

where R_t is the set of provenance roots behind the target hypothesis's evidence. If everything known descends from one dataset, independent evidence is maximally valuable.

Tractability. $\mathrm{Tr}(a) = (1 - \mathrm{time}_a) \cdot \rho^m$, where m is the number of required resources not declared available and ρ (default 0.5) is the missing-resource penalty. With no declared resource set, all resources are assumed available.

Cost. The action's configured relative cost, passed through.

Calibration. Information gain is measured in bits and is typically well below 1 for a single action, while the other terms are $[0, 1]$ scores; the default configuration therefore sets $w_{\mathrm{ig}} = 10$ to place the terms on a comparable scale, with all weights exposed in YAML.

3.6 Planning

The planner is deliberately thin: it scores every candidate with the injected utility estimator, sorts by total utility, truncates to top-K, and attaches to each recommendation its utility breakdown, its target hypothesis, and a reasoning string naming the dominant term. All ranking policy lives in the (replaceable) utility estimator.

3.7 Belief Updating

New evidence arrives as a likelihood vector $L = \big(P(e \mid h_1), \ldots, P(e \mid h_n)\big)$; hypotheses absent from the vector receive a neutral likelihood of 1. The default updater is exact discrete Bayes,

$$p_i' \;=\; \frac{p_i\, P(e \mid h_i)}{\sum_j p_j\, P(e \mid h_j)}, \tag{5}$$

with two safeguards: evidence impossible under every hypothesis leaves the distribution unchanged (it is uninformative for this hypothesis set), and the update is non-destructive, recording the evidence identifier on each hypothesis's supporting or contradicting list according to the direction of the probability change. The updater is a protocol; richer models (soft evidence, correlated hypotheses, hierarchical priors) can be substituted without touching callers.

4 Empirical Grounding from Historical Outcomes (Paper 3)

The utility decomposition of Section 3.5 was derived from first principles. Paper 3's predictor-discovery study now lets us check those principles against a thousand historical outcomes: 980 questions frozen at five cutoffs, each labeled by a strict blinded judge with what the following five years of literature actually did (52.7% substantively addressed; interpretable models predict 2020 outcomes at AUC 0.71 out-of-time and out-of-domain). Four of its findings bear directly on this framework.

Explicit competing hypotheses are the one falsifiability property with predictive teeth. Across controlled analyses, explicitly articulated competing hypotheses raise the probability of future attention ($\beta = +0.17$, $p = 0.02$) and, directionally, of being *answered* given attention ($\beta = 0.56$, $p = 0.08$) — the only falsifiability feature that survives Paper 3's four-way robustness screen. This is direct empirical support for the hypothesis-decomposition step this framework makes mandatory: a question is planned *as* a set of competing hypotheses, never as free text.

The resolution channel is discriminative, not diffuse. Within Paper 3's tension-mined subset (374 questions with verbatim evidence-graph provenance), every premise refutation comes from a *quantified* tension cluster (3/63 quantified vs. 0/311 unquantified), five of six resolutions are explicit-contradiction-typed, all six resolutions come from tensions *recurrent* across cutoffs — and clusters built on methodological-challenge language engaged the community at the average rate yet resolved *nothing* (0/61). Translated into this framework's terms: evidence with a high likelihood ratio against a specific, contestable claim is what closes questions; diffuse methodological worry produces attention without resolution. This is precisely the behavior encoded in the information-gain estimator's discriminative-power table, and the reason our utility centers expected entropy reduction rather than engagement.

Attention is community-relational; novelty is not the signal. Paper 3's robust predictors of future attention are prior community recognition (overlap with pre-cutoff reviews, $\beta = +0.34$), the density of directly related prior evidence ($\beta = +0.26$), and cluster breadth within the tension subset ($\beta = +0.45/\mathrm{SD}$) — while dropping the semantic-novelty feature group *improves* prediction, and instrument diversity of the existing evidence base carries a negative controlled effect ($\beta = -0.21$): multi-instrument questions read as synthesis-level problems no single group owns. Five of six pre-stated mechanism hypotheses, including "evidence tension beats novelty", failed. For this framework the lesson is a separation of objectives: our planner optimizes expected *resolution* (entropy reduction), not expected attention, and its novelty term is provenance independence — the step from one evidence root to a second, which Paper 3's refutation cases (quantified cross-checks of a single-source claim) support — not semantic distance from the literature, which Paper 3 shows predicts nothing. We accordingly ship an empirically calibrated configuration (`configs/paper3_informed.yaml`) that keeps the information-gain weight, halves the novelty weight, and raises tractability (Paper 3's Cox model: archival-data availability accelerates engagement; requiring a not-yet-operational instrument is the largest negative effect). Section 6 shows its effect on the case-study ranking.

A shared benchmark substrate. Paper 3 releases its questions, provenance clusters, and tiered outcome labels. Its records load directly through our question loader (an importer ships in `experiments/`), and each labeled question is a candidate benchmark unit for Section 5: the cutoff supplies the historical-simulation boundary and the label supplies hidden ground truth. Its learned structure prior (`results/structure_prior.json`, structure-only AUC 0.586 vs. 0.502 for the full question-level record within a single source) is likewise importable as an empirical prior over which evidence configurations the community will engage — a community-uptake signal complementary to our resolution-centric utility.

5 Offline Evaluation by Historical Simulation

An offline planner can be evaluated the only way that avoids self-grading: against history. The benchmark takes a *full* timestamped evidence graph and a cutoff τ:

1. Filter the graph to nodes with timestamp $< \tau$ (untimestamped structural nodes are kept; edges touching hidden nodes are dropped). The planner sees only this sub-graph.
2. Run any planner pipeline (a two-method protocol) on the visible sub-graph, producing recommendations and the hypothesis set used.
3. Express post-cutoff evidence as ground-truth records: which hypothesis it bore on, which action type historically produced it, and its per-hypothesis likelihoods.
4. A ground-truth item is *reached* if some recommendation matches its action type and target hypothesis; reached items are replayed through the belief updater to measure the uncertainty the plan could have removed.

The central question is: *did the planner recommend the actions that would have resolved the question earlier?*

5.1 Metrics

Top-K utility. Mean total utility of the top-K recommendations.

Hypothesis entropy reduction. $H(p_{\text{before}}) - H(p_{\text{after}})$ after replaying reached ground truth — what history says the plan was worth.

Average information gain. Mean *expected* gain across recommendations — what the planner believed; comparing it with realized entropy reduction exposes over- or under-confidence.

Action diversity. Distinct action types over number of recommendations, penalizing degenerate plans.

Coverage of competing hypotheses. Fraction of hypotheses targeted by at least one recommendation; a good plan probes the space of explanations, not just the front-runner.

Resolution efficiency. Fraction of ground-truth evidence reached by the plan; 1.0 means every piece of evidence that historically resolved the question was on the recommended path.

Expected utility. Rank-discounted utility with weights 1/rank (normalized), modeling that earlier recommendations are likelier to be executed.

6 Case Study

We instantiate the framework on a question from exoplanet atmospheric characterization — *"Is the reported low water abundance real or a retrieval artifact?"* — stressing that the domain lives entirely in the data files; the code is untouched. The question is not arbitrary: it is the question family of Paper 2's pilot benchmark, whose documented history (carried forward into Paper 3's dataset) records that the reported strongly subsolar water abundance was a premise *later refuted by three independent re-analyses* — exactly the convergence the question called for. Our hidden ground truth follows that history: three post-cutoff evidence items (two independent retrieval re-analyses and one cross-dataset comparison, 2020–2021) that support the retrieval-artifact hypothesis and contradict the "real" one.

Four hypotheses compete: the low abundance is real ($p = 0.3$); it is a retrieval artifact ($p = 0.3$); it is a cloud–composition degeneracy ($p = 0.2$); it is an instrument systematic ($p = 0.2$). The pre-cutoff graph (cutoff $\tau = 2020$; the planner sees 10 of 16 nodes) contains the discovery paper and an early re-analysis, both derived from the *same* transmission spectrum dataset — textbook provenance concentration.

Gap analysis flags the "real" hypothesis for lacking independent corroboration (all evidence descends from one dataset), the "artifact" hypothesis for sparse evidence, and the remaining two for having no evidence. Under the default configuration the planner's top recommendation is re-running the retrieval under alternative settings, targeting the artifact hypothesis — the action family that historically refuted the premise — followed by independent observation and cross-dataset comparison for the two leading hypotheses.

Table 1 summarizes the run. Resolution efficiency of 1.0 indicates every piece of historically decisive evidence was on the recommended path; replaying the three refuting items shifts the belief mass decisively onto the artifact hypothesis, removing 1.06 of the 1.97 bits of initial hypothesis entropy. Coverage of 0.5 reflects that the two low-prior hypotheses did not make the top-5 under default weights — raising the coverage-relevant weights or K trades focus for breadth, and the benchmark makes that trade-off measurable.

Re-ranking the same candidates under the Paper 3–informed configuration of Section 4 sharpens the match with history further: the discriminating re-analysis stays first, cross-dataset comparison rises to second and third, and the expensive new observation — which history shows was *not* the

Table 1: Benchmark results for the case study (top-K=5, cutoff 2020, default configuration).

Metric	Value
Top-K utility	2.596
Hypothesis entropy reduction (bits)	1.062
Average information gain (bits)	0.173
Action diversity	0.600
Coverage of competing hypotheses	0.500
Resolution efficiency	1.000
Expected utility	2.616

resolving step for this question — drops to the bottom of the top-5. An empirical prior learned from a thousand historical outcomes and a utility derived from first principles agree on the head of the ranking and disagree exactly where Paper 3's evidence says they should: on how much raw source novelty is worth.

The utility weights also behave as designed: with all weights at 1, cheap archive searches dominate the ranking and the decisive (expensive) observation is never recommended; the calibrated default ($w_{\mathrm{ig}} = 10$) restores the balance between bits and $[0, 1]$ scores. We report this deliberately — the failure mode is exactly the kind of miscalibration the benchmark exists to expose.

7 Scaling Up: 374 Historical Tension Questions

The case study is one fully specified benchmark unit. To turn the metrics into distributions, we batch-convert every evidence-tension question in Paper 3's released dataset — 374 questions across eight astronomy subfields and five cutoffs, each storing its generating evidence cluster verbatim (sentences, arXiv ids, publication dates; 244 four-paper, 59 three-paper, 71 two-paper clusters) together with a blinded future-outcome label and cluster-level structure flags.

Conversion. Each cluster becomes an evidence graph by transparent, auditable rules (implemented in `experiments/paper3_benchmark.py`). The four template hypotheses receive priors seeded from the cluster's structure flags: quantified tensions (scaled by their N-sigma) shift mass toward "real effect", contradiction and method-challenge language toward "analysis artifact", detection-vs-limit stance opposition toward "unmodeled confounder", and cross-facility clusters toward "measurement systematic". Every cluster sentence becomes a claim node derived from its (timestamped) paper node, with support edges assigned by the same flags; hypotheses whose explanatory channel has no flag remain evidence-free, which is exactly what gap analysis exists to surface. Distinct arXiv ids count as distinct provenance roots — an upper bound on independence, since shared underlying datasets are not visible at this granularity.

Results. The planner runs the full pipeline at each question's own historical cutoff. Table 2 reports the distribution of every at-scale-computable metric. Resolution efficiency is deliberately absent: Paper 3's labels record *whether and when* a question was addressed, not *through which action type*, so the matching that metric requires is not derivable at scale — mining action types from the resolving papers is the concrete next step. For the premise-refuted cases the label does fix the direction of the hidden evidence; all three trace to a single physical case (the FRB Galactic-latitude

Table 2: Metric distributions over the 374 tension questions (top-K=5, default configuration, each question at its own cutoff).

Metric	Mean	Std	Median	IQR
Top-K utility	1.904	0.146	1.923	[1.902, 1.923]
Average information gain (bits)	0.056	0.007	0.056	[0.054, 0.056]
Action diversity	0.604	0.131	0.600	[0.600, 0.600]
Coverage of competing hypotheses	0.505	0.110	0.500	[0.500, 0.500]
Expected utility	1.932	0.147	1.952	[1.925, 1.952]

discrepancy, independently re-mined at three successive cutoffs), for which the direction-only replay removes 0.095 bits each.

Planner utility is a neglect signal. Stratifying by Paper 3's outcome label yields the run's most instructive result: questions the community subsequently *addressed* score systematically *lower* planner utility than questions it did not (top-K utility 1.887 vs. 1.919, seeded permutation test $p = 0.004$; expected utility 1.916 vs. 1.946, $p = 0.004$; hypothesis coverage 0.488 vs. 0.520, $p = 0.007$). The mechanism is visible in the conversion: later-addressed questions have richer evidence structure (quantified, explicitly contradictory clusters — precisely Paper 3's attention predictors), which yields denser graphs, fewer under-evidenced hypotheses, and therefore fewer high-utility open actions. The planner assigns its highest scores to exactly the evidence-poor configurations the community subsequently neglected. This is the planner-side face of Paper 3's central finding that attention follows preparation rather than open gaps — and it is the intended behavior, not a defect: a tool that allocates evidence-acquisition effort by expected resolution will, by construction, point away from where the community is already converging and toward where uncertainty is largest. The two allocation policies are complementary, and the benchmark now measures their divergence quantitatively.

Scope notes. The distributions are narrower than a mature benchmark should produce, because 2–4-sentence clusters support only schematic graphs; full-text evidence extraction would widen the spread. The conversion rules are heuristics operating on cluster-level flags, not per-sentence stance; a stance-aware mapping is the other obvious refinement. Both limitations are properties of the imported data's granularity, not of the benchmark protocol.

8 Implementation

The reference implementation is pure Python (standard library plus PyYAML), organized as ten modules that communicate only through shared dataclasses and the graph interface; estimators and generators are injected, never imported across modules. All tunable behavior — utility weights, gap thresholds, the action catalog, discriminative powers, source novelties, resource sets — lives in YAML or documented constructor parameters. The test suite (48 unit tests) covers question loading, the graph, gap analysis, action generation, every utility component, planner ranking, belief updating, each evaluation metric, and the end-to-end benchmark. An experiment script reproduces Section 6 in one command, and a second importer script loads Paper 3's released question and outcome-label records directly into planner objects (their extra fields — source, cutoff, provenance cluster — are preserved in question metadata).

9 Limitations and Future Work

- **Two-outcome information gain.** The preposterior model collapses an action's result to favors/disfavors with a per-type likelihood ratio. Multi-outcome models, and ratios conditioned on the specific hypothesis pair, are natural extensions behind the same interface.
- **Type-and-target ground-truth matching.** Resolution efficiency currently matches recommendations to hidden evidence by action type and target hypothesis; semantic matching would credit plans that reach the same evidence by a different route.
- **Exhaustive, exclusive hypothesis sets.** The Bayesian updater assumes the hypothesis set partitions the possibility space; an explicit "none of the above" mass is a small but valuable addition.
- **Static candidate space.** Actions are instantiated from gaps in the current graph; sequential planning over multi-step campaigns (with evidence-dependent branching) is the obvious next step, and the offline design makes it benchmarkable the same way.
- **Benchmark granularity.** The scaled run (Section 7) covers 374 questions, but its graphs are schematic — built from 2–4 abstract sentences via cluster-level flags, without per-sentence stance — and its labels lack the action types needed for resolution efficiency at scale. The two concrete refinements are stance-aware edge mapping and mining action types from the papers that historically resolved each question; only the fully specified case study currently exercises all seven metrics.
- **Empirical grounding is correlational and one-domain.** Paper 3's effects come from astronomy alone, from abstract-level retrieval, and from an LLM judge with measured (low) inter-rater ceilings; the calibrated configuration inherits those caveats. Its own prospective instance (frozen 2026-08, scored 2027–2030) will test the imported prior on a future no model has seen.

10 Conclusion

We presented an offline, domain-independent framework that turns "what should we investigate next?" into a concrete, configurable, and — crucially — historically evaluable planning problem. By separating evidence representation, gap analysis, action generation, utility estimation, and belief updating behind small interfaces, the framework serves as a reference implementation that individual research programs can adapt by swapping components and editing configuration, not code. The historical benchmark reframes evaluation from plausibility to counterfactual usefulness: a good evidence planner is one that would have gotten science to the answer sooner.

The connection to Paper 3 closes a loop the series has been building toward. Paper 3 shows that what the community *will* engage is predictable and community-relational, while what *resolves* questions is discriminative — quantified, contradictory, recurrent evidence. This framework operationalizes the second finding as a planning objective, imports the first as a calibrated configuration, and on the series' own canonical case study recommends, first, the exact action family that history used to settle the question. Run at scale over all 374 of Paper 3's tension questions, the planner's utility turns out to anti-correlate with subsequent community attention: it prizes precisely the evidence-poor questions the community left behind. That is the division of labor made measurable — communities follow preparation, a resolution-driven planner points at neglect — and it suggests the most useful deployment is not predicting what science will do, but surfacing what science is skipping. Testing that allocation against Paper 3's pre-registered prospective instance (frozen 2026, scored 2027–2030) is the next experiment, and the infrastructure for it is now in place.